



Amharic Document Representation for Adhoc Retrieval

Tilahun Yeshambel, Josiane Mothe, Yaregal Assabie

► To cite this version:

Tilahun Yeshambel, Josiane Mothe, Yaregal Assabie. Amharic Document Representation for Adhoc Retrieval. KDIR 2020, Nov 2020, Online conference, Hungary. hal-02960435

HAL Id: hal-02960435

<https://hal.science/hal-02960435v1>

Submitted on 7 Oct 2020

HAL is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.

Amharic Document Representation for Adhoc Retrieval

Tilahun Yeshambel¹, Josiane Mothe² and Yaregal Assabie³

¹*IT PhD program, Addis Ababa University, Addis Ababa, Ethiopia*

²*INSPE, Univ.de Toulouse, IRIT, UMR5505 CNRS, Toulouse, France*

³*Department of Computer Science, Addis Ababa University, Addis Ababa, Ethiopia*

tilahun.yeshambel@uog.edu.et, josiane.mothe@irit.fr, yaregal.assabie@aaau.edu.et

Keywords: Adhoc Retrieval, Amharic, Complex Morphology, Stem, Root.

Abstract: Amharic is the official language of the government of Ethiopia currently having an estimated population of over 110 million. Like other Semitic languages, Amharic is characterized by complex morphology where thousands of words are generated from a single root form through inflection and derivation. This has made the development of tools for Amharic natural language processing a non-trivial task. Amharic adhoc retrieval faces difficulties due to the complex morphological structure of the language. In this paper, the impact of morphological features on the representation of Amharic documents and queries for adhoc retrieval is investigated. We analyze the effects of stem-based and root-based approaches on Amharic adhoc retrieval effectiveness. Various experiments are conducted on TREC-like Amharic information retrieval test collection using standard evaluation framework and measures. The findings show that a root-based approach outperforms the conventional stem-based approach that prevails in many other languages.

1 INTRODUCTION

Searching digital information on the Web or document collection has long become part of the human daily life. Information Retrieval (IR) is the task of searching relevant documents to a user query from document collection. Both the research community and the industry have been very active in this field for more than 60 years (Sanderson and Croft, 2012). Nowadays, IR has gained much attention due to the explosion of digital data and the need of accessing relevant information from huge corpus quickly and accurately.

IR systems work based on documents representing natural languages, and consequently, the characteristics of a given language affects the whole process of IR (Moukdad, 2002). Thus, natural language processing (NLP) has attracted the attention of IR community since a long time (Smeaton, 1992; Jackson and Moulinier, 2007; Cambria and White, 2014). For example, NLP applications and resources provide a means to find better representative terms for indexing and query terms that improve search results. This calls for the need of dealing with language specific issues to improve the performance of IR systems. The morphology, orthography, tokenization, syntax,

semantics, and compound splitting of a language are some of the issues to be considered while developing IR systems. It has long been understood that linguistic variation has significant impact on IR effectiveness as it leads to the omission of relevant documents to users' queries (Moukdad, 2002). Many languages have different forms generated from a single word due to morphology and orthography. Identifying the basic units of words is more difficult for morphologically complex languages than for simple languages. Performing simple matching between words generated from the same root is not applicable to capture similarity.

Thus, in order to come up with an effective IR system, one has to deal with the complex characteristics of the language. One of the key features of Amharic is its complex morphology which itself leads to complex grammatical structure. As a result, finding better representations for documents and queries has been an issue of theoretical discussion in Amharic IR. The forms that can be considered for document and query representation are stems and roots. In many languages, the use of surface forms of words to represent documents and queries is not taken into consideration due to the proliferation of words that can be generated from a single root form. This issue

is imperative in the development Amharic IR. Although some efforts have been made to develop Amharic IR systems using stems, their effectiveness with respect to the use of various forms has not been systematically analyzed thus far. Therefore, this research analyzes the use of stems and roots for content representation and investigates their effects on Amharic IR.

The rest of this paper is organized as follows. Section 2 describes Amharic language and its morphology. Section 3 discusses related work and Section 4 presents how documents and queries are represented in Amharic IR system. Experimental results and evaluation are discussed in Section 5. In Section 6, we make conclusion along with the way forward in Amharic IR.

2 AMHARIC LANGUAGE

Amharic is the official language of the government of Ethiopia. Although several languages are spoken in Ethiopia, Amharic is spoken as a mother tongue by a sizeable proportion of the country's population currently estimated to be over 110 million. Among the Semitic language family, it is the second most spoken language in the world, next to Arabic. Due to its historical significance and official status, Amharic has been serving as the *lingua franca* of the country since a long time. As a result, many literary works, government documents, educational materials, religious literary works, etc. are predominantly produced in Amharic. Amharic uses Ethiopic script for writing having 34 base characters (with a vowel አ /ə/), each of which are modified to have six other orders representing vowels in the order of ኡ /u/, ኢ /i/, ኣ /a/, ኤ /e/, ኦ /i/, and ኦ /o/.

Like other Semitic languages, complex morphological processes are carried out on Amharic word classes such as verbs, nouns and adjectives (Yimam, 2001). Amharic verbs are the most complex word classes and can be generated by attaching affixes on verbal stems. On the other hand, verbal stems can be generated from verbal roots by inserting vowels between radicals. For example, the verbal stem ገደል-/gədäl-/ is derived from the verbal root ገ-ደ-ል/g-d-l/. Moreover, verbal stems (e.g. ተገደል-/təgədäl-/) can be derived from other verbal stems (e.g. ገደል-/gədäl-/) by affixing morphemes. The verb formation process is usually completed by attaching a verbal stem with person, gender, number, case, tense/aspect and mood markers. For example, from the verbal stem ገደል-/gədäl-/ the following verbs can be generated: ገደልኩ/gədäлку'I killed',

ገደልኩህ/gədäлкуh'I killed you', ገደልን/gədäln'we kill', ተገደልኩ/təgədäлку'I was killed', ገደለች/gədälät/'she killed', etc. As verbs are marked for subject and object, they alone can represent a complete sentence. For example, the word አልሰበረንም/əlsəbərənīm'he did not break us', which is constructed from the morphemes ገል-səbər-ə-ni-m, is a complete sentence with the following linguistic information: ገል-...-m /not/, -səbər- /did break/, -ə-/he/ and -ni-/us/. Accordingly, thousands of verbs can be derived from a verbal root through a complex morphological process carried out by attaching a combination of person, case, gender, number, tense, aspect, mood and others (Abate and Assabie, 2014; Assabie, 2017).

Based on a morphological structure, Amharic nouns and adjectives can be either derived or non-derived. For example, the word መሬት/məret'earth' and ዛፍ/zaft'tree' are non-derived nouns whereas words like ስብራት /sibirat'the state of being broken' and ደኅነት/dəginät 'generosity' are nouns derived from the verbal root ስ-ብ-ር/s-b-r 'to break' and the adjective ደኅ/dəg 'generous', respectively. Derived nouns are generated from other word classes through morphological processes. In general, Amharic nouns can be derived from verbal roots, adjectives and other nouns by affixing vowels or bound morphemes. Derived adjectives can be formed from verbal roots by infixing vowels between consonants (e.g. ከ-ብ-ደ/k-b-d 'to become heavy' → ከባድ/kəbad 'heavy'), nouns by suffixing bound morphemes such as -ኝ /əña/ (e.g. ገልበት/gulbat 'power' → ገልበተኛ/gulbatəña 'powerful') and verbal stems by prefixing or suffixing bound morphemes (e.g. ደካም-/dəkam-/ → ደካማ/dəkama 'weak'). Although the morphological process of derivation of nouns and adjectives is complex by itself, even more complexity arises from their inflections. Amharic nouns and adjectives are inflected for number by suffixing -ች/-ot/ or -ች/-wot/, definiteness by suffixing -ኡ/-əu/ or -ዉ/-wu/, objective case by suffixing -ን/-n/, possessive case by suffixing different morphemes depending on the subject, and gender by suffixing -ኢት/-it/. These inflections can appear alone or in combination at the same time, along with prepositions and negation markers which lead to the generation of thousands of word forms from a single noun or adjective. For example, ያለባለቤቶቹ/jaləbaləbetot'u 'without the owners of the house' is generated from the morphemes ገል-ገል-balə-bet-ot/-u (ገ preposition 'of/with', ገል negation marker 'not/without', balə possessive marker 'owner of', bet noun 'house', ot/ plural marker, and u definite

marker 'the') where the core morpheme is the noun *bet* 'house'.

3 RELATED WORK

Semitic languages are known to pose unique challenges in the development of NLP applications due to their complex morphologies. These challenges are propagated to the development of IR systems since the effectiveness of IR systems depends on the availability of various NLP tools and resources. In this section, we discuss the techniques and NLP resources used to develop IR systems for Semitic languages in general.

Arabic is the largest of the Semitic language family. Arabic IR systems have a long history (Ambati *et al.*, 2008; Larkey *et al.*, 2007; Darwish and Magdy, 2014). For example, Al-Hadid *et al.* (2014) developed a neural network-based model where documents and queries are represented using stems and their similarity is computed using cosine similarity. Musaid (2000) investigated the effectiveness of word-based, stem-based, and root-based representation of documents and queries. The word-based and stem-based approaches miss relevant documents while root-based approach retrieves non-relevant documents. Moukdad (2002) conducted a research to compare the effects of stem and root on Arabic IR. The retrieval effectiveness of stem and root were evaluated on search engine. The results of the experiments indicate stemming is more effective than root. Larkey *et al.* (2007) investigated the effects of light stemming (removal of prefix and suffix) on Arabic IR. A comparison between stem-based and root-based retrieval was performed. The finding indicates light stemming outperforms than root and other stemmers which are based on detail morphological analysis. Abdusalam (2008) presented an Arabic text retrieval technique using lexicon-based light stemming. The study evaluated the effectiveness of lexicon-based light stemming, Arabic patterns, root, expanding query and filtering foreign words using n-grams. According to the results, the preprocessing techniques like normalization, stopword removal and light-stemming improve retrieval results whereas n-grams and roots decrease the performance. The lexicon-based stemming and the relevance feedback approaches perform better than light-stemming approach alone. Ali *et al.* (2020) investigated the effect of morphological analysis on Arabic IR. A rule-based stemmer was used to extract the root/stem of words to be used as indexing and searching terms.

The results showed slight improvement on IR effectiveness due to the stemmer.

Hebrew is one of the Semitic languages spoken mainly in Israel. Carmel and Maarek (1999) presented a morphological disambiguator based on a statistical approach that takes advantage of an existing morphological analyzer. The approach is context-free and was used for query analysis and linguistic indexing of text documents. Instead of words, the morphological patterns were used for disambiguation. The statistical morphological disambiguator returns only the best base form(s), or lemma(s). It makes the decisions of the most likely set of analyses based on the frequency of the morphological patterns associated with the analyses of the input word. The disambiguator was tested by integrating with the Hebrew search engine. It conflates all inflectional forms and the performance of the search engine increased. Ornan (2002) designed Hebrew search engine by applying a rule-based morphological analysis. The design of the search engine takes into account the construction of a morphological, syntactic and semantics analyzer. The search engine eliminates words unsuited both to the syntax and the semantic of a sentence.

Although Amharic is significantly used in Ethiopia, the status of IR system development for the language is relatively at rudimentary level. Alemayehu and Willett (2003) studied the retrieval effectiveness of word-based, stem-based, and root-based approaches on Amharic language. The experiments were carried out by running 40 queries on 548 documents using OKAPI system and the study concludes that stem-based retrieval is slightly better than root-based. Similarly, Mindaye *et al.* (2010) developed an Amharic search engine using stems. The system was tested with 11 queries on 75 news documents. The average precision and recall values were 0.65 and 0.95, respectively using OR operator in between query terms, and 0.99 and 0.52, respectively for AND operator. Argaw *et al.* (2004) developed dictionary-based Amharic-English IR system. Documents and queries were represented using Bag-Of-Words (BOW). Stopwords were removed using Inverse Document Frequency (IDF) and stopwords list. The average precisions of 0.3615 and 0.4009 were achieved using IDF and stopwords list, respectively. Argaw *et al.* (2006) build dictionary-based Amharic-French IR system with and without word sense discrimination using BOW approach. Stemming was applied to remove prefix and suffix. The experiments were conducted on SICS and Lucene search engines. Stopwords were removed by using IDF. The result of SICS is better

than Lucene. The word sense discrimination performs slightly better than non discrimination.

While there are several studies that focused on the development of IR systems for Semitic languages, most of them have followed the techniques employed for morphologically simple languages like English. This has not produced the desired retrieval result as documents could not be represented appropriately. Only few studies have tried to consider the issue of document representation in a systematic way. In our approach, we address this crucial issue of document representation in the development of Amharic IR.

4 DESIGN OF AMHARIC IR

The main objective of this work is to systematically identify the optimal representations for documents (and queries) in Amharic IR. It focuses on the selection of the structures of terms and stopwords based on the morphological characteristics of the language. Taking these issues into account, we also propose an Amharic retrieval system which is slightly different from the basic architecture of IR systems. In our case, stopwords are removed after the application of morphological analysis on documents and queries as shown in Figure 1. Both documents and queries pass through the same preprocessing tasks that involve language specific tokenization, character normalization, and removal of punctuation marks. Text preprocessing is followed by morphological analysis which is performed on documents and queries to select appropriate terms for document representation. Morphological analysis

is among the key tasks in our IR system as it helps to remove stopwords from documents and queries.

4.1 Preprocessing

Preprocessing includes tags removal, tokenization, character normalization and punctuation mark removal. Tokenization is done using space and punctuation marks as delimiters of words. Character normalization is made to represent various characters having similar pronunciation using a single grapheme. Base characters having such property are {ሀ /ha/, ሐ /ha/, ኀ /ha/ and ኸ /ha/}, {ሠ /sa/ and ሰ /sa/}, {ጸ /ts'a/ and ፀ /ts'a/}, and {ኣ /pa/ and ዐ /pa/}. Furthermore, the fourth order characters {ሃ /ha/, ሓ /ha/, ኃ /ha/ and ከ /ha/} and {ኣ /pa/ and ዓ /pa/} have similarity phonemes with the correspondence base character. Therefore, Amharic character normalization involves mapping of ሀ /ha/, ሃ /ha/, ሐ /ha/, ሓ /ha/, ኀ /ha/, ኃ /ha/, ኸ /ha/, ከ /ha/, ዐ /pa/, and ዓ /pa/ to ኣ /pa/, etc. For example, the word ስለጸሀፊዋ can also be written as ሥለጸሀፊዋ, ስለጸሐፊዋ, ሥለጸሐፊዋ, ሥለጸኃፊዋ, ስለጸኃፊዋ, ሥለጸሐፊዋ, ስለፀሀፊዋ, ስለፀሐፊዋ, ሥለፀኃፊዋ, ስለፀኃፊዋ, etc. although some of them rarely appear in a text.

4.2 Morphological Analysis

Documents and user information need should be represented appropriately using terms that will be used later for matching query with document. It is to be noted that indexing terms are weighted based on the word frequency. In IR, most often, the variants of a word are conflated during indexing into a single form. It has the advantage of making the calculation of indexing term frequency straightforward.

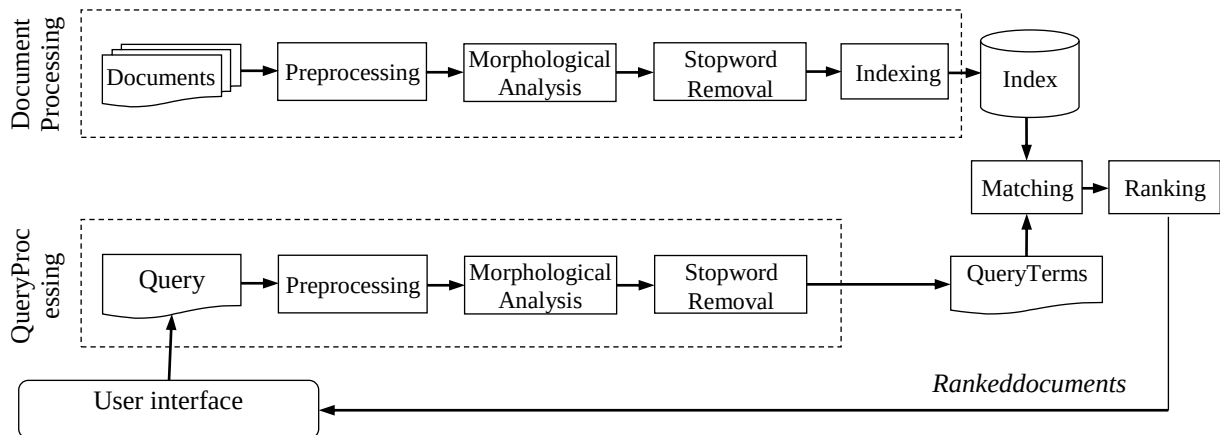


Figure 1: Basic architecture of the proposed Amharic IR system.

Therefore, in this research, we study the feasibility of stem-based and root-based document representation with respect to their effectiveness for Amharic IR. Since well-designed Amharic morphological analyzer is not available, we design semi-automatic annotation to morphologically segment words in documents.

4.2.1 Stem-Based Morphological Analysis

One of the bases of Amharic words are stems. A large number of words are formed by attaching affixes to stems. Therefore, morphological analysis should be carried out to extract the stem from the rest of morphemes. For example, the morphological structures of the primary noun በመንገድቻችን/bəməngədocat/in'by our roads'/and the adjective የደጋጎች/jədəgagot/'of generous'/are shown as follows.

በመንገድቻችን	የደጋጎች
በ_መንገድ_ኦች_አችን	የ_ደጋ_ኦች
pre-stem-pl-1,pl ¹	gen-stem-pl
by-road-many-our	of-generous-many

Similarly, variants of adjectives and nouns derived from primary nouns are mapped into their common stems of nouns. For example, the derived noun ስለልጅነቴና/siləlidzinətena'and about my childhood'/ and the derived adjective ዓለማዊ/ʔələməwi'worldly'/ are morphologically segmented as follows.

ስለልጅነቴና	ዓለማዊ
ስለ_ልጅ_ነት_ኤ_ና	ዓለም_አዊ
pre-stem-nom-1,s-con	sem-adj
about-child-being-my-and	world-suf

Amharic verbs undergo complex morphological process. Verbs are marked for person, gender, number, tense, subject, object, and negation by attaching a series of affixes. For example, the word ፈለገችኛሁ/fəlagət/ət/ihu'she wanted you'/and the verb ከአልተስማማናቸውም/kəʔəltəsmamanat/əwim'and if we are not comfortable for them'/ is analysed as follows.

ፈለገችኛሁ	ከአልተስማማናቸውም
ፈለግ_ኸች_ኦችሁ	ከ_አል_ስማም_ኃ_አቸው_ም

¹ 1: first person, 2: second person, 3: third person, s: singular, pl: plural, pre: preposition, suf: suffix, nom: nominative, con: conjunction, neg: negative, gen: genitive, def: definite marker, adj: adjectivizer, sub: subject, obj: object

stem-sub-obj	pre-neg-stem-1,pl-3,pl-neg
search-she-you	from-not-comfort-we-they-not

Amharic has both basic and derived (causative, passive, infinitive and reduplicative) types of verbal stems. Causative stems are formed using the prefixes አ-/ʔə-/ and አስ-/ʔəs-/ whereas passive stems are formed using the prefix ተ-/tə-/. Infinitive stems are also formed using the prefix መ-/mə-/ and reduplicative stems are formed by duplicating the middle consonant. For example, the words ከተሰበረ/kətasəbərə'if it is broken'/ and ሰበረኝሁ/səbabərat/ihu'you repeatedly broke'/ have the derived stems ተሰበር-/təsəbər-/ and ሰበር-/səbər-/, respectively, but a common basic stem ሰበር-/səbər-/. From the semantics point of view, there is no conceptual difference between derived and basic stems. Moreover, the derived stems are generated from basic stems which are common forms for many variants than derived stems. Therefore, in the case of stem-based indexing and retrieval, variants of a word are represented using their basic stems.

Stemming is usually applied since lemmatization is more computationally consuming for just slight effectiveness improvements (Balakrishnan and Lloyd-Yemoh, 2014). Stemming has also been applied in Amharic IR systems (Mindaye et al., 2010; Alemayehu and Willett, 2003). However, stemming is not expected to produce the desired result of term frequency in Amharic text. Morphological variants of Amharic verbs can have more than one stem. For example, morphological variants such as ሰበረ/səbərə'he broke', ተሰበረ/təsəbari'broken', and አሰበረ/ʔəsəbərə'he helped to break' have the basic stems ሰበር-/səbər-/, ሰበር-/səbar-/, and ሰበር-/səbər-/, respectively. As a result, stemming provides distorted frequency since each stem of variants is counted differently though they are semantically similar. Therefore, Amharic verbal stems need one more reduction analysis to extract root. Indeed, verbal stems are themselves formed from roots.

4.2.2 Root-Based Morphological Analysis

Roots are the basis for the formation of basic stems and many other variants of the same Amharic word. Though some words are derived from stems, their origins are roots rather than stems. For example, the morphological structures of the verb ከሰበረኝሁ/kəsəbərək'at/əw'if I break them', the derived noun ስብራቴ/sibirate'my broken belonging'/ are as follows.

ከሰበረኝሁ	ስብራቴ
--------	------

ከ_ሰበር_ከ_አቸው	ስብር_አት_ኤ
pre-stem-1,s-3,pl	stem-nom-1s
ከ_ስብር_ከ_አቸው	ስብር_አት_ኤ
pre-root-1,ps-3,pl	root-nom-1ps
from-break-I-them	break-nom-my

In the above example, stems of the two words have different forms while they have the same root i.e., ስብር/s-b-r/. According to Yimam (2001), more than 10 basic stems can be generated from a given root. As depicted in Table 1, variants of words describe similar concept, but they have different stems. This affects the term frequency which has an impact on ranking and retrieval of documents. On the other hand, all variants have a single root. Therefore, root-based representation maps morphologically related words into one common form. Accordingly, statistics information based on root-based approach can be computed accurately so that the actual term frequency can be known. The actual number of variants will be equal to the frequency of their common root. However, the frequency of each stem will be less than the actual occurrence of variants. Moreover, root form increases the matching possibilities between query terms and index terms. Thus, the root-based approach becomes a better way to represent documents and queries for Amharic IR. We have also experimentally analyzed the viabilities of stem-based and root-based approaches.

4.3 Stopword Removal

Stopwords are words that evenly occur in many documents and serve as purpose rather than content. Thus, they are removed from documents and query. Stopwords can be removed either by applying a list or IDF. In morphologically simple languages

like English, stopword identification and removal is achieved by considering a list of words that are identified to be stopwords. The conventional trend applied so far for removing Amharic stopwords is also to use a list. However, taking the characteristics of language into consideration, this is certainly not the most appropriate way. Indeed, Amharic stopwords: (i) do not necessarily exist as standalone words; (ii) can accept prefixes and suffixes; and (iii) may exist as part of Amharic words and serve as prefix or suffix. For these reasons, it is not possible to find and remove all Amharic stopwords unless the morphological structure of words is known. For example, one may consider words like "the" a stopword in English. Its Amharic equivalent is a suffix "-ኡ/-ሁ/" or "-ው/-ሠ/" that does not appear as a standalone word. Accordingly, "the house" and "the student", for instance, are equivalent to ቤት /betu/ (ቤት /bet/ + -ኡ/-ሁ/) and ተማሪው /tāmariw/ (ተማሪ /tāmari/ + -ው/-ሠ/), respectively. As there could be several sequences of affixes representing articles, prepositions, numbers, etc., words can appear in various morphological structures. It means that one could not work with surface forms of words to identify and remove stopwords as most of the stopwords in Amharic do not exist as standalone words. This indicates that stopword identification and term representation in Amharic IR demands a different consideration than the conventional trend.

Yeshambelet *et al.* (2020a) constructed root-based and stem-based Amharic stopword lists by considering the semantics of Amharic words and corpus statistics. The values of frequency, variance, entropy and mean in a large corpus were used while constructing the stopword list.

Table 1: Amharic root and stems of variants.

Root	Basic stems	Variants	Concept
ት-ል-ቅ	ትልቅ	ትልቁ, ትልቃችን, ትልቃቸው, ትልልቅ, etc.	big
	ታላቅ	ታላቁ, የታላቁ, ከታላቁ, ታላላቆቹ, ታላቅ, etc.	
	ተለቅ	ተለቁ, ተለቆች, ተለቁ, ተለቅን, ተለቃችሁ, etc.	
ምስክር	መስከር	መስከረ, መስከረች, መስከሩ, ተመስከረ, etc.	witness
	መሳከር	ተመሳከረ, አመሳከረ, ይመሳከራል, etc.	
	መሳከር	እናመሳከር, አመሳከር, አመሳከሪ, etc.	
	መስከር	ይመስከር, እንመስከር, ልመስከር, etc.	
	መስከር	እንመስከራለን, ይመስከራሉ, ትመስከራለች, etc.	

In both cases, it is shown that stopwords significantly impact on retrieval effectiveness, size of index, and term weighting of non-stopwords. Experimental results also showed that the root-based approach is better than the stem-based approach in conflating all variants of a stopword. The identified stopwords include prepositions (e.g. *ወደ/wədə'to'*, *ስለ/silə'about'*, *እስከ/ʔiskə'up to'*, *ብ-/bə-'by'*, *ከ-/kə-'from'*, etc.), conjunctions (e.g. *እና/ʔina'and'*, *ይሁንአንጅ/yihunʔindʒi'however'*, *እዚህ/ʔizih'here'*, etc.), negation markers (*አል...ም/ʔəl...m'not'*), indefinite articles (*አንድ/ʔənd'an'*), auxiliary verbs (*አል-/ʔ-l'say'*, *ገ-ብ-ር/ገ-b-r'was'*, etc.), *ወዘተ/wəzətə'and so on'*, etc.

4.4 Indexing

To test the effect of morphological analysis on Amharic IR, stem-based and root-based indexes are created using Lemur toolkit. Lemur is also used for retrieval purpose. The stem-based index was created using the basic stems of words while the root-based index was created using the root of words.

4.5 Matching

In the proposed system, document processing involves text preprocessing, morphological analysis, stopword removal and indexing. As a result of this process, we obtain indexed documents. On the query side, we apply similar processes except indexing. Thus, query processing provides a set of terms representing information needs of users. Searching of relevant documents is carried out by matching query terms (representing information need of users) with index terms (representing documents). In this work, we use exact vocabulary term matching. It searches documents that contain the query terms without analysing the semantics of words and without considering the semantic connections between them.

4.6 Ranking

For a given query Q and the collection of retrieved documents D, the Lemur toolkit ranked retrieval results based on their relevance. The document length and number of matching query terms are considered. OKAPI ranks documents based on the following algorithm.

$$\text{Score}(D, Q) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot (1 - b + b \cdot \frac{|D|}{\text{avgdl}})} \quad (1)$$

Where $f(q_i, D)$ is q_i 's term frequency in the document D, $|D|$ is the length of the document D in words, and avgdl is the average document length in the text collection from which documents are drawn. k_1 and b are free parameters. In this research the value of $k_1 = 2.0$ and $b = 0.75$. $\text{IDF}(q_i)$ is the IDF (inverse document frequency) weight of the query term q_i .

In case of LM, the KL divergence ranking function with default parameters was used. It captures the term occurrence distributions and computed as follow.

$$\sum_{w: c(w, d) > 0, p(w|\theta Q) > 0} p(w|\theta Q) \log \frac{ps(w|d)}{\alpha p(w|C)} + \log \alpha d \quad (2)$$

Where d is document, w is word, $p(w|\theta Q)$ is a query model, is estimated query, $p(w|C)$ is the collection language model, $ps(w|d)$ is the smoothed probability of a word seen in the document, αd is a coefficient controlling.

5 EXPERIMENT

5.1 Experimental Data

The experimental setup followed the format of the Text Retrieval Conference (TREC). Experiments were carried out using a scientifically built corpus (Yeshambelet *al.*, 2020b) and a stopword list constructed by Yeshambelet *al.* (2020a). The test collection has 12,538 documents and 240 queries while the stopword list contains 222 stopwords.

5.2 Implementation and Measures

Python was used for the preprocessing tasks while indexing and retrieval were performed using Lemur toolkit, which is a search engine designed to support research on language model (LM) for IR tasks². The retrieval effectiveness was evaluated automatically using *trec_eval* tool which can compute many evaluation measures³. LM and BM25 models were used as retrieval models.

²<http://www.lemurproject.org>

³http://trec.nist.gov/trec_eval

- የአየርንብረትብክለትመንስኤወች /jəʔəjərnibrətbiklət mənʃiʔewət/;

Figure 3: Recall-Precision curves based on stems and roots.

Comparison with Google Amharic retrieval

The Google Amharic search engine is based on stem. It returns different retrieval results in different ranks for the same query using different variants of query words. Similar results are obtained in our work while we apply stem-based approach. For example, Google search results of the queries የአጥንት ስብራት/jəʔət'intsibirat'being broken bone' and የአጥንት መስበር/jəʔət'intməṣəbər'the process of being broken bone' are different though the same concept is expressed via different variants. Our approach differs from Google search engine into two ways.

- i. Google searches based on both basic stems and derived stems. It returns different retrieval results for basic stems and derived stems queries though they are semantically similar. However, in our work, the stem-based approach is based on basic stems only, providing the same retrieval results for both basic stems and derived stems.
- ii. Google does not employ roots to represent verbs and words derived from them. However, we use root-based approach as it conflates all variants of words to a common form.

Comparison of LM and BM25

We also compared the performance of LM and BM25. Although LM is very popular and powerful for IR in different languages, previous Amharic IR researches were made based on classical IR models such as vector space model (Mindayeet *al.* 2010; Argawet *al.*, 2004; Argawet *al.*, 2006). Language modelling was not employed for Amharic IR. In this study, the impact of language modelling retrieval on Amharic IR is also investigated. As depicted in Figure 4, the blue line representing LM is above the red line representing BM25. Both precision and recall values of LM are better than that of BM25 at different levels. This is because of the capability of LM to capture the dependency of words and estimate the probability distribution of a query in each document. This means LM is a more suitable retrieval model for Amharic language. The roots of verbs and words derived from them and the basic-stems of other word classes are robust to represent words not only in IR but also in other applications such as machine translation, information extraction, sentiment analysis, etc.

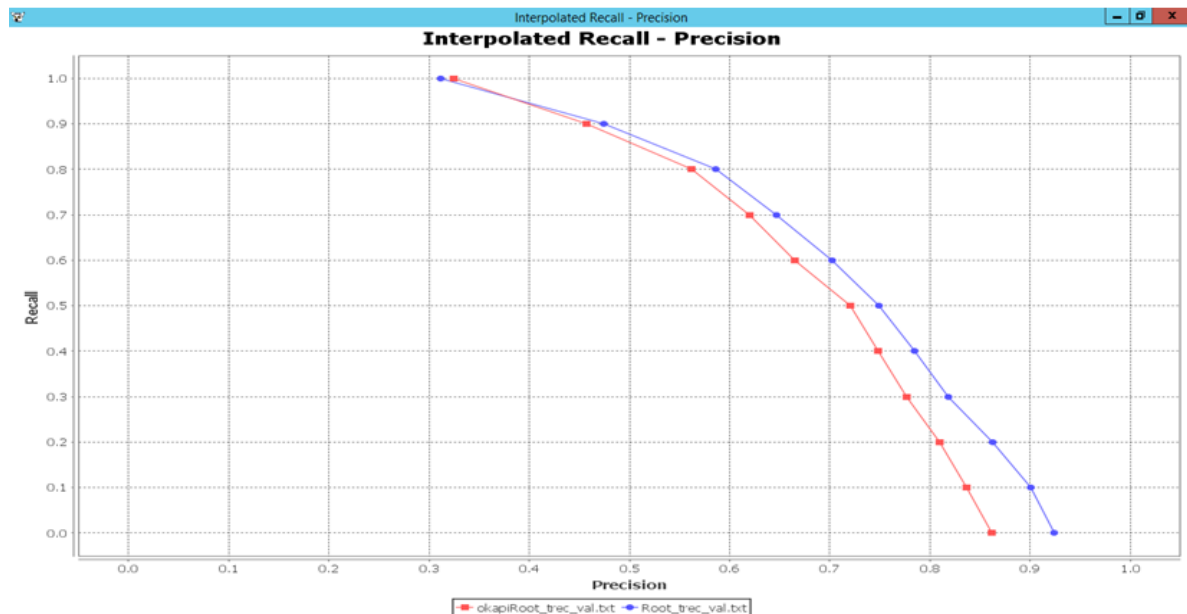


Figure 4: Comparison of LM and BM25 retrieval models on root-based approach.

6 CONCLUSION

The development of Amharic IR demands thorough investigation of the characteristics of the language. Its complex morphology affects the way documents and queries are represented for the task of information retrieval. In this work, we conducted several experiments using various forms of words for document representation. Experimental results have shown that root forms of words provide better results in representing documents. It is also shown that, with the use of root forms, the LM retrieval is better than BM25 model. As the proposed system is based on exact vocabulary term matching, future work needs to consider query expansion so as to take into account synonyms, collocation words, name identification, etc. Furthermore, the language has many ambiguous words which have different meaning in various contexts. Thus, this work may be improved by handling ambiguity.

REFERENCES

- Abate, M. and Assabie, Y., 2014. Development of Amharic morphological analyzer using memorybased learning. In *Proc. of the 9th Int. Conf. on Natural Language Processing*, pp. 1-13, Warsaw.
- Abdusalam, A., 2008. Effective retrieval techniques for Arabic text. *PhD Thesis*, School of Computer Science and Information Technology, RMIT University, Melbourne, Victoria, Australia.
- Alemayehu, N. and Willett, P., 2003. The effectiveness of stemming for information retrieval in Amharic. *Program: Electronic Library and Information Systems*, 37(4), pp.254–259.
- Al-Hadid, I., Afaneh, S., Al-Tarawneh, H., and Al-Malahmeh, H., 2014. Arabic information retrieval system using the neural network model. *International Journal of Advanced Research in Computer and Communication Engineering*, Vol. 3, Issue 12.
- Ali, A., Mosa, E., and Abdullah, B., 2020. An intelligent use of stemmer and morphology analysis for Arabic information retrieval. *Egyptian Informatics Journal*, <https://doi.org/10.1016/j.eij.2020.02.004>.
- Ambati, V., Rohini, U., Pramod, P., Balakrishnan, N., and Reddy, R., 2006. Multilingual information access: Information Retrieval and Translation in a Digital Library. In: *Proc. of the 2nd Int. Conf. on Universal Digital Library*, Alexandria, Egypt.
- Argaw, A. A., Asker, L., Cöster, R., and Karlgren, J., 2004. Dictionary-based Amharic-English information retrieval. In *Workshop of the Cross-Language Evaluation Forum for European Languages*, pp. 143–149, Springer.
- Argaw, A. A., Asker, L., Cöster, R., Karlgren, J., and Sahlgren, M., 2006. Dictionary-based Amharic-French information retrieval. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 4022 LNCS, 83–92. https://doi.org/10.1007/11878773_9.
- Assabie, Y., 2017. Development of Amharic morphological analyzer. *Technical Report*. Ethiopian Ministry of Communication and Information Technology, Addis Ababa, Ethiopia.
- Balakrishnan, V., and Lloyd-Yemoh, E., 2014. Stemming and Lemmatization: A Comparison of Retrieval Performances. *Lecture Notes on Software Engineering*, 2(3), 262–267. <https://doi.org/10.7763/lnse.2014.v2.134>.
- Cambria, E. and White, B., 2014. Jumping NLP curves: A review of natural language processing research, *IEEE Computational Intelligence Magazine*, 9(2): 48-57.
- Carmel, D. and Maarek, Y., 1999. Morphological disambiguation for Hebrew search systems. *International Workshop on Next Generation Information Technologies and Systems; NGITS 1999: Next Generation Information Technologies and Systems*, pp. 312-325.
- Darwish, K. and Magdy, W., 2014. Arabic information retrieval. *Foundations and Trends in Information Retrieval*, 7(4): 239-342.
- Jackson, P. and Moulinier I., 2007. Natural language processing for online applications: *Text retrieval, extraction and categorization*. John Benjamins Publishing, Amsterdam, Netherlands, 2nd ed.
- Larkey, L. S., Ballesteros, L., and Connell, M. E. 2007. Light stemming for Arabic information retrieval. In *Arabic Computational Morphology*, pp.221-243. Springer, Dordrecht.
- Mindaye, T., Redewan, H. and Atnafu, S., 2010. Design and implementation of Amharic search engine. In *Proc. of the 5th Int. Conf. on Signal Image Technology and Internet Based Systems*, pp. 318–325.
- Munye, M. and Atnafu, S., 2012. Amharic-English bilingual web search engine. *Proceedings of the International Conference on Management of Emergent Digital EcoSystems, MEDES 2012*, pp. 32–39.
- Moukdad, H., 2002. A comparison of root and stemming techniques for the retrieval of Arabic documents.

- PhD Thesis*, Graduate School of Library and Information Studies, McGill University, Montreal.
- Musaid, S., 2000. Arabic information retrieval system-based on morphological analysis (AIRSMA): A comparative study of word, stem, root and morpho-semantic methods. *PhD Thesis*, Computer and Information Science, DeMontfort University, United Kingdom.
- Ornan, U., 2002. A morphological, syntactic and semantic search engine for Hebrew texts. In: *Proceedings of the ACL-02 Workshop on Computational Approaches to Semitic Languages*. Philadelphia, Pennsylvania, USA.
- Sanderson, M., and Croft, W., 2012. The history of information retrieval research. In *Proceedings of the IEEE*, Special Centennial Issue, pp.1444-1451.
- Smeaton, A. F., 1992. Progress in the application of natural language processing to information retrieval tasks. *The Computer Journal*, 35(3), pp. 268-278.
- Yeshambel, T., Mothe, J. and Assabie, Y., 2020a. Construction of morpheme-based Amharic stopwordlist for information retrieval system. In: *Proc. of ICAST2020*, Springer.
- Yeshambel, T., Mothe, J. and Assabie, Y., 2020b. 2AIRC: The Amharic Adhoc information retrieval test collection. In: *Proc. of CLEF2020*, LNCS 12260, pp. 55-66.
- Yimam, B., 2001. *Yamarignasewasiw* (Amharic grammar). CASE. Addis Ababa, 2nd edition.